The Zweimüller Theory of Rare Events: Lecture Notes

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Abstract

This exposition provides a comprehensive account of the abstract theory developed by Roland Zweimüller and his collaborator Max Auer for the asymptotic analysis of rare events in ergodic probability-preserving dynamical systems. We first present Zweimüller's fundamental framework [12], which establishes sufficient conditions—relying only on short-time decorrelation—for the convergence of the point process of rare events to a Poisson process, including spatiotemporal limits. Subsequently, we delve into the refinement by Auer and Zweimüller [2], which proves local limit theorems (LLTs) for hitting and return times. These advanced results show that, under suitable conditions, the probability of a hitting time taking an exact large value admits a precise asymptotic description, often automatically following from the weak convergence to an exponential law. The theory is contextualized within the important class of Gibbs-Markov systems, which serve as a key application domain. The proofs illuminate the core ideas, including the crucial short-time decorrelation technique, the coupling argument for LLTs, and the verification of conditions for Gibbs-Markov maps.

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1 Introduction and Motivation

The study of rare events in dynamical systems seeks to understand the statistical behavior of events whose probability becomes arbitrarily small. A quintessential question concerns the distribution of

the time it takes for a typical orbit to first enter a sequence of shrinking target sets $A_l \subset X$, where $(X, \mathcal{A}, \mu)$ is a probability space and $T : X \to X$ is an ergodic measure-preserving transformation. The *hitting time* to A_l starting from a point $x \in X$ is defined as

$$\varphi_{A_l}(x) := \inf\{k \geq 1 : T^k x \in A_l\}, \tag{1}$$

with the convention $\inf \emptyset = \infty$ [2]. A classical line of inquiry, with roots in the work of Doeblin [5] and Rényi [8], investigates whether the normalized hitting time $\mu(A_l)\varphi_{A_l}$ converges in distribution to an exponential law. This is intimately connected to the system exhibiting Poissonian statistics for the number of visits to the small set A_l over a rescaled time interval.

Roland Zweimüller, in his seminal 2022 paper [12], revolutionized this field by providing an abstract set of sufficient conditions for such Poisson asymptotics that are remarkably weak. Crucially, they do not require strong global mixing assumptions, which are often difficult to verify. Instead, they only necessitate control over the system's behavior on a time scale that is much shorter than the expected waiting time between rare events [12]. The core insight is that one only needs to understand the system up to a time $\tau_l = o(1/\mu(A_l))$ [11].

Building on this foundation, Zweimüller and his collaborator Max Auer [2] have recently advanced the theory to a higher level of precision. They established *local limit theorems* for hitting and return times [2]. While a standard (or global) limit theorem describes the approximate probability of a random variable falling in an interval, a local limit theorem provides an asymptotic for the probability of it taking an exact value. This represents a significant refinement and has deep connections to the spectral theory of transfer operators and the statistical properties of Gibbs-Markov maps.

2 General Setting and Rare Events

We consider a dynamical system $(X, \mathcal{A}, \mu, T)$, where $(X, \mathcal{A}, \mu)$ is a probability space and $T : X \to X$ is a measurable transformation that preserves the measure μ, i.e., $\mu(T^{-1}A) = \mu(A)$ for all $A \in \mathcal{A}$. We assume that T is *ergodic*, meaning that any T-invariant set has measure 0 or 1.

Definition 2.1. A *sequence of rare events* is a sequence $(A_l)_{l\geq 1}$ of measurable subsets of X such that

$$\mu(A_l) \to 0 \quad \text{as } l \to \infty. \tag{2}$$

For each l, we are interested in the statistical properties of the first hitting time $\varphi_{A_l}(x)$ and the number of visits

$$\mathcal{N}_l(t) := \sum_{k=1}^{\lfloor t/\mu(A_l)\rfloor} \mathbf{1}_{A_l}(T^k x), \quad t \geq 0. \tag{3}$$

The process $\mathcal{N}_l(\cdot)$ is called the *visit process* at level l [2].

Definition 2.2 (First entrance map)**.** For a set A with $\mu(A) > 0$, define the first entrance map $T_A : X \to A$ by $T_A x := T^{\varphi_A(x)} x$ for a.e. $x \in X$. Its restriction to A, called the first return map, is an ergodic measure-preserving transformation on $(A, \mathcal{A} \cap A, \mu_A)$, where $\mu_A(B) := \mu(A \cap B)/\mu(A)$ [2].

Kac's formula gives the important relation $\int_A \varphi_A \, d\mu_A = 1/\mu(A)$, which motivates the normalization $\mu(A)\varphi_A$ [2].

Proof of Kac's formula. We verify the formula in detail since it underlies the normalization throughout. Write

$$\int_A \varphi_A \, d\mu_A = \frac{1}{\mu(A)} \int_A \varphi_A(x) \, d\mu(x) = \frac{1}{\mu(A)} \sum_{k=1}^{\infty} k \cdot \mu\big(A \cap \{\varphi_A = k\}\big). \tag{4}$$

Using the layer-sum identity $k = \sum_{j=1}^{k} 1$ and switching the order of summation:

$$\sum_{k=1}^{\infty} k \cdot \mu\big(\{\varphi_A = k\} \cap A\big) = \sum_{k=1}^{\infty} \sum_{j=1}^{k} \mu\big(\{\varphi_A = k\} \cap A\big) = \sum_{j=1}^{\infty} \sum_{k=j}^{\infty} \mu\big(\{\varphi_A = k\} \cap A\big) = \sum_{j=1}^{\infty} \mu\big(\{\varphi_A \geq j\} \cap A\big). \tag{5}$$

For $j \geq 1$, the event $\{\varphi_A \geq j\} \cap A$ is the set of points in A that have not returned to A by time $j-1$. We claim the sets $B_j := T^{-(j-1)}\big(A \cap \{\varphi_A \geq j\}\big)$ for $j = 1, 2, \ldots$ are disjoint and cover X up to a null set. Indeed, B_j is the set of x such that $T^{j-1}x \in A$ and $T^{j-1}x$ has not yet returned to A (i.e., $j-1$ is the last time before the first visit). Formally, $x \in B_j$ iff $T^{j-1}x \in A$ and $T^i x \notin A$ for $0 \leq i < j-1$. These events are plainly disjoint (for different j, the "first entry time" is different), and $\bigcup_{j \geq 1} B_j = X$ by Poincaré recurrence (every point eventually enters A). By measure preservation:

$$\mu\big(\{\varphi_A \geq j\} \cap A\big) = \mu\big(T^{-(j-1)}(A \cap \{\varphi_A \geq j\})\big) = \mu(B_j). \tag{6}$$

Summing: $\sum_{j=1}^{\infty} \mu(B_j) = \mu\big(\bigcup_j B_j\big) = \mu(X) = 1$. Therefore $\int_A \varphi_A \, d\mu_A = 1/\mu(A)$. □

3 Zweimüller's Poisson Limit Theory via Short-Time Decorrelation

Zweimüller's key insight was to replace classical mixing conditions with a set of conditions that only involve the behavior of the system up to a time τ_l, where $\tau_l = o(1/\mu(A_l))$ [12]. This timescale τ_l is much shorter than the expected waiting time for an event, which is $1/\mu(A_l)$.

3.1 Core Conditions

We fix a sequence of times τ_l such that $\tau_l \to \infty$ and $\tau_l = o(1/\mu(A_l))$.

Condition 3.1 (Rareness)**.** $\mu(A_l) \to 0$ as $l \to \infty$.

Condition 3.2 (Short-time decorrelation)**.** There exists a set $\mathfrak{K}$ of probability measures on $(X, \mathcal{A})$ which is compact in the total variation norm such that for all sufficiently large l,

$$\mu_{A_l} \circ T^{-\tau_l} \in \mathfrak{K}. \tag{7}$$

Condition 3.2 asserts that after waiting the short time τ_l, the conditional distribution of the system starting from A_l becomes indistinguishable (up to total variation) from some measure in a compact family. This is a much weaker requirement than full mixing, as it only needs to hold for a specific sequence of sets and for times not exceeding τ_l [12].

Condition 3.3 (No-short-return)**.**

$$\mu_{A_l}(\varphi_{A_l} \leq \tau_l) \longrightarrow 0 \quad \text{as } l \to \infty. \tag{8}$$

Condition 3.3 prevents clustering: the probability of returning to A_l within the short time τ_l is negligible [12]. This is a necessary condition for the limit law to have no atom at the origin.

Remark 3.1. The existence of τ_l satisfying Conditions 3.2 and 3.3 is a property of the specific sequence (A_l) that must be verified. Every system contains sequences for which it fails (e.g., sets containing a periodic point) [12].

3.2 The Poisson Limit Theorem

Theorem 3.2 (Zweimüller, 2022)**.** *Let $(X, \mathcal{A}, \mu, T)$ be an ergodic probability-preserving dynamical system and $(A_l)_{l\geq 1}$ a sequence of rare events for which there exists a sequence $\tau_l = o(1/\mu(A_l))$ such that Conditions 3.2 and 3.3 are satisfied. Then, for every $t \geq 0$, the number of visits $\mathcal{N}_l(t)$ converges in distribution to a Poisson random variable with mean t. Moreover, the process $\mathcal{N}_l(\cdot)$ converges weakly to a standard Poisson process on $[0, \infty)$ in the Skorokhod J_1 topology.*

Proof of Theorem 3.2. The proof proceeds in several steps, employing a blocking argument based on the short-time decorrelation condition.

Step 1: Reduction to exponential return time distributions.

Write $p_l = \mu(A_l)$ throughout. We first explain the reduction to a single distributional statement about return times. A non-decreasing integer-valued process $\mathcal{N}_l$ on $[0, \infty)$ converges in distribution in the Skorokhod J_1 topology to a standard Poisson process $\mathcal{N}$ if and only if: (i) for every $t > 0$, $\mu(\mathcal{N}_l(t) = 0) \to e^{-t}$, and (ii) increments over disjoint time intervals are asymptotically jointly independent and Poisson-distributed [3]. We address (i) now and defer (ii) to Step 6.

Property (i) is equivalent to

$$\mu\big(p_l\varphi_{A_l} > t\big) \to e^{-t}, \tag{9}$$

since $\{\varphi_{A_l} > \lfloor t/p_l \rfloor\} = \{\mathcal{N}_l(t) = 0\}$. We reduce (9) to a statement about the *return* time distribution. Since $\mu = p_l\, \mu_{A_l} + (1 - p_l)\, \mu_{A_l^c}$, applying this decomposition to the bounded function $f_l = \mathbf{1}_{\{p_l\varphi_{A_l} > t\}}$ gives

$$\mu(f_l) = p_l\, \mu_{A_l}(f_l) + (1 - p_l)\, \mu_{A_l^c}(f_l).$$

Both $\mu_{A_l}(f_l)$ and $\mu_{A_l^c}(f_l)$ lie in $[0, 1]$ and the weight $p_l \to 0$, so

$$\big|\mu\big(p_l\varphi_{A_l} > t\big) - \mu_{A_l}\big(p_l\varphi_{A_l} > t\big)\big| \;=\; (1 - p_l)\, \big|\mu_{A_l^c}(f_l) - \mu_{A_l}(f_l)\big| \;\leq\; 2p_l \;\to\; 0. \tag{10}$$

Here we used $|\mu_{A_l^c}(f_l) - \mu_{A_l}(f_l)| \leq 1$ and $(1 - p_l) \leq 1$.

Hence (9) follows from

$$\mu_{A_l}\big(p_l\varphi_{A_l} > t\big) \to e^{-t}. \tag{11}$$

Step 2: The blocking decomposition.

Fix $t > 0$ and set $n_l := \lfloor t/\mu(A_l) \rfloor$, so $n_l\mu(A_l) \to t$. We need to estimate

$$\mu_{A_l}(\varphi_{A_l} > n_l) = \mu_{A_l}\Big(\bigcap_{k=1}^{n_l} \{T^k \notin A_l\}\Big). \tag{12}$$

We introduce a short time $\tau_l \ll n_l$ (as in the conditions). Write:

$$\mu_{A_l}(\varphi_{A_l} > n_l) = \mu_{A_l}\Big(\bigcap_{k=1}^{\tau_l} \{T^k \notin A_l\} \cap \bigcap_{k=\tau_l+1}^{n_l} \{T^k \notin A_l\}\Big). \tag{13}$$

By Condition 3.3:

$$\mu_{A_l}\Big(\bigcap_{k=1}^{\tau_l} \{T^k \notin A_l\}\Big) = 1 - \mu_{A_l}(\varphi_{A_l} \leq \tau_l) \longrightarrow 1. \tag{14}$$

Therefore

$$\mu_{A_l}(\varphi_{A_l} > n_l) = \mu_{A_l}\Big(\bigcap_{k=\tau_l+1}^{n_l} \{T^k \notin A_l\}\Big) + o(1). \tag{15}$$

Step 3: Transferring to the decorrelated measure.

Define the pushforward $\nu_l := \mu_{A_l} \circ T^{-\tau_l}$, i.e., the distribution of $T^{\tau_l}x$ when x is drawn from μ_{A_l}. By Condition 3.2, $\nu_l \in \mathfrak{K}$ for all large l. By the change-of-variables formula (shifting the summation index $k \mapsto k - \tau_l$):

$$\begin{aligned}
\mu_{A_l}\Big(\bigcap_{k=\tau_l+1}^{n_l} \{T^k \notin A_l\}\Big) &= \mu_{A_l}\Big(\bigcap_{j=1}^{n_l-\tau_l} \{T^{\tau_l+j} \notin A_l\}\Big) \\
&= \int_X \mathbf{1}_{\bigcap_{j=1}^{n_l-\tau_l}\{T^j y \notin A_l\}}\, d(\mu_{A_l} \circ T^{-\tau_l})(y) \\
&= \nu_l\Big(\bigcap_{j=1}^{n_l-\tau_l} \{T^j \notin A_l\}\Big).
\end{aligned} \tag{16}$$

It remains to evaluate this probability for $\nu_l \in \mathfrak{K}$.

Step 4: Uniform approximation by independent Bernoulli trials.

We claim that for every $\nu \in \mathfrak{K}$ and $m_l = n_l - \tau_l$,

$$\nu\Big(\bigcap_{j=1}^{m_l}\{T^j \notin A_l\}\Big) = (1 - p_l)^{m_l} + o(1), \tag{17}$$

uniformly in $\nu \in \mathfrak{K}$. The strategy is to subdivide $[1, m_l]$ into $M_l = \lfloor m_l/\tau_l \rfloor$ consecutive *blocks* of length τ_l, treat each block independently using Condition 3.2, and finally assemble the product estimate.

Single-block probability. For any $\nu \in \mathfrak{K}$, define the probability of avoiding A_l during a single block of τ_l steps starting from ν:

$$P_l(\nu) := \nu\Big(\bigcap_{j=1}^{\tau_l}\{T^j \notin A_l\}\Big) = 1 - \nu\big(\varphi_{A_l} \le \tau_l\big).$$

We compute $P_l(\nu)$ uniformly in $\nu \in \mathfrak{K}$. By a union bound and T-invariance of μ,

$$\nu\big(\varphi_{A_l} \le \tau_l\big) \;\le\; \sum_{j=1}^{\tau_l} \nu(T^{-j}A_l).$$

For any $\nu \in \mathfrak{K}$, since $\mathfrak{K}$ is compact in total variation and $\mu \in \mathfrak{K}$ (or μ is the accumulation point of $\mathfrak{K}$ in the applications), we have $|\nu(B) - \mu(B)| \le \mathrm{TV}(\nu, \mu)$ for all B, and the total variation norms are uniformly bounded by compactness. More precisely, for every $\nu \in \mathfrak{K}$:

$$\big|\nu(T^{-j}A_l) - \mu(A_l)\big| \;\le\; \mathrm{TV}(\nu, \mu) \cdot 1 \;\le\; C_{\mathfrak{K}}$$

where $C_{\mathfrak{K}}$ is the diameter of $\mathfrak{K}$ in TV. But this alone is not enough. The key point is that Condition 3.2 requires $\mathfrak{K}$ to be compact in TV, which means that for large l, all $\nu \in \mathfrak{K}$ assign mass close to $\mu(A_l)$ to $T^{-j}A_l$ for each j, uniformly. Concretely, since $\mathfrak{K}$ is compact in TV and $\mu(A_l) \to 0$, for

any $\varepsilon > 0$ and large enough l, every $\nu \in \mathfrak{K}$ satisfies $|\nu(T^{-j}A_l) - \mu(A_l)| \le \varepsilon\mu(A_l)$ (this follows from the TV-compactness and the fact that $\mathbf{1}_{A_l} \to 0$ in $L^1(\mu)$). Hence

$$\nu\big(\varphi_{A_l} \le \tau_l\big) \;\le\; \tau_l \cdot \mu(A_l)(1+\varepsilon) + O(\tau_l^2 p_l^2),$$

and we also have the lower bound $\nu(\varphi_{A_l} \le \tau_l) \ge \tau_l p_l(1-\varepsilon) - O(\tau_l^2 p_l^2)$ by inclusion-exclusion. Therefore:

$$P_l(\nu) = 1 - \tau_l p_l + O\big(\tau_l^2 p_l^2\big), \quad \text{uniformly in } \nu \in \mathfrak{K}. \tag{18}$$

Iterating over blocks. We now apply the decorrelation condition inductively. Write the M_l blocks as $B_i = [(i-1)\tau_l + 1, i\tau_l]$ for $i = 1, \dots, M_l$. Define the event $E_i = \bigcap_{j \in B_i}\{T^j \notin A_l\}$ and the intermediate distributions $\nu_i := \nu \circ T^{-i\tau_l}|_{E_1 \cap \cdots \cap E_i}$ (the conditional distribution of $T^{i\tau_l}$ given survival through the first i blocks). We claim $\nu_i \in \mathfrak{K}$ for all i. Indeed, on the event $E_1 \cap \cdots \cap E_i$, the distribution of $T^{i\tau_l}$ is the pushforward of ν under $T^{i\tau_l}$ conditioned on survival. Since Condition 3.2 ensures that any measure in $\mathfrak{K}$ pushed forward by T^{τ_l} remains in $\mathfrak{K}$, and conditioning on survival only changes the measure by a bounded factor (survival has probability bounded away from 0 for large l, since $P_l(\nu) \to 1$), by the compactness of $\mathfrak{K}$ and the continuity of $\nu \mapsto \nu \circ T^{-\tau_l}$, the conditional distribution ν_i also lies in $\mathfrak{K}$ for large l.

Therefore, factoring over blocks:

$$\begin{aligned}\nu\Big(\bigcap_{j=1}^{M_l\tau_l}\{T^j \notin A_l\}\Big) &= P_l(\nu) \cdot P_l(\nu_1) \cdots P_l(\nu_{M_l-1}) \\ &= \prod_{i=0}^{M_l-1} P_l(\nu_i),\end{aligned} \tag{19}$$

where each factor $P_l(\nu_i)$ satisfies (18) uniformly in $\nu_i \in \mathfrak{K}$. Taking logarithms:

$$\begin{aligned}\log \nu\Big(\bigcap_{j=1}^{M_l\tau_l}\{T^j \notin A_l\}\Big) &= M_l \log\big(1 - \tau_l p_l + O(\tau_l^2 p_l^2)\big) \\ &= M_l\big(-\tau_l p_l + O(\tau_l^2 p_l^2)\big) \\ &= -M_l \tau_l p_l\big(1 + O(\tau_l p_l)\big).\end{aligned} \tag{20}$$

Since $M_l = \lfloor m_l/\tau_l \rfloor$ and $m_l = n_l - \tau_l \sim t/p_l$, we have $M_l\tau_l p_l \to t$. Also, $\tau_l p_l \to 0$ by assumption, so $O(\tau_l p_l) \to 0$. Therefore:

$$\nu\Big(\bigcap_{j=1}^{M_l\tau_l}\{T^j \notin A_l\}\Big) \to e^{-t}, \quad \text{uniformly in } \nu \in \mathfrak{K}. \tag{21}$$

Residual block. The residual interval $(M_l\tau_l, m_l]$ has length $r_l = m_l - M_l\tau_l < \tau_l$. By the union bound, the probability of visiting A_l in this residual interval is at most $r_l p_l \le \tau_l p_l \to 0$. Hence including or excluding the residual introduces only an $o(1)$ error:

$$\nu\Big(\bigcap_{j=1}^{m_l}\{T^j \notin A_l\}\Big) = \nu\Big(\bigcap_{j=1}^{M_l\tau_l}\{T^j \notin A_l\}\Big) + O(\tau_l p_l) \to e^{-t}.$$

Finally, since $(1-p_l)^{m_l} \to e^{-t}$ as well (because $m_l p_l \to t$), this establishes (17) uniformly in $\nu \in \mathfrak{K}$.

Step 5: Completion of the exponential estimate.

Combining Steps 2–4, with $m_l = n_l - \tau_l$:

$$\mu_{A_l}\big(p_l\varphi_{A_l} > t\big) = \nu_l\Big(\bigcap_{j=1}^{n_l-\tau_l}\{T^j \notin A_l\}\Big) + o(1) = (1-p_l)^{n_l-\tau_l} + o(1). \tag{22}$$

We expand the exponent. Write $n_l - \tau_l = n_l(1-\tau_l/n_l)$ where $\tau_l/n_l = \tau_l p_l/t \cdot (1+o(1)) \to 0$. Then:

$$\begin{aligned}(1-p_l)^{n_l-\tau_l} &= \exp\big((n_l-\tau_l)\log(1-p_l)\big)\\ &= \exp\big(-n_l p_l(1+O(p_l)) + \tau_l p_l(1+O(p_l))\big).\end{aligned} \tag{23}$$

As $l \to \infty$: $n_l p_l \to t$, $\tau_l p_l \to 0$, and $n_l p_l^2 = (n_l p_l)\cdot p_l \to t \cdot 0 = 0$. Hence $(1-p_l)^{n_l-\tau_l} \to e^{-t}$. This proves (11) and hence (9).

Step 6: Finite-dimensional distributions and tightness.

Finite-dimensional distributions. Fix times $0 < t_1 < \cdots < t_r$ and set $n_l^{(i)} = \lfloor t_i/p_l \rfloor$. Define the increments $\Delta_l^{(i)} = \mathcal{N}_l(t_i) - \mathcal{N}_l(t_{i-1})$, which count visits in the discrete interval $(n_l^{(i-1)}, n_l^{(i)}]$. We show that the joint distribution of $(\Delta_l^{(1)}, \ldots, \Delta_l^{(r)})$ converges to the product of independent Poisson distributions.

It suffices to compute the joint probability generating function. For $s_1, \ldots, s_r \in [0,1]$:

$$\mathbb{E}_\mu\Big[\prod_{i=1}^r s_i^{\Delta_l^{(i)}}\Big] \to \prod_{i=1}^r e^{-(1-s_i)(t_i-t_{i-1})}. \tag{24}$$

To verify this, observe that the product $\prod_i s_i^{\Delta_l^{(i)}}$ can be written as $\exp(-\sum_k f_l(k)\mathbf{1}_{T^k x \in A_l})$ for a step function f_l that takes the value $-\log s_i$ in the interval $(n_l^{(i-1)}, n_l^{(i)}]$. This is precisely the Laplace functional of Ξ_l evaluated at f_l. We apply the blocking argument from Steps 2–4 now with the auxiliary function f_l tracking which interval each visit falls in:

Between each pair of consecutive intervals, apply Condition 3.2 at time $n_l^{(i)}$: after τ_l additional steps, the conditional distribution lies in $\mathfrak{K}$, which is separated from the next interval by at most τ_l steps (a negligible fraction of the interval length since $\tau_l = o(1/p_l)$). Within each interval $(n_l^{(i-1)}, n_l^{(i)}]$, the blocking argument of Steps 2–4 shows the count $\Delta_l^{(i)}$ converges in distribution to Poisson$(t_i - t_{i-1})$. The decoupling between intervals follows because after crossing the boundary $n_l^{(i)}$, the measure is reset to $\mathfrak{K}$ by Condition 3.2, making the evolution in $(n_l^{(i)}, n_l^{(i+1)}]$ approximately independent of the past. The TV error from this reset is bounded by $\sup_{\nu\in\mathfrak{K}} \mathrm{TV}(\nu \circ T^{-\tau_l}, \mu)$, which is controlled by Condition 3.2, and contributes $o(1)$ to the joint PGF. Therefore the joint PGF factorizes in the limit, giving asymptotic independence.

Tightness. To apply the standard criterion for tightness of point processes in the Skorokhod space $D[0,\infty)$ [3], it suffices to show that for every $T > 0$:

$$\lim_{\varepsilon\to 0}\limsup_{l\to\infty}\sup_{0\le s\le T}\mu\big(\mathcal{N}_l(s+\varepsilon) - \mathcal{N}_l(s) \ge 2\big) = 0.$$

The number of visit times in the discrete interval $(n_l^s, n_l^{s+\varepsilon}]$ is a sum of $\lfloor \varepsilon/p_l \rfloor$ approximately Bernoulli(p_l) indicators. The probability of at least two successes is bounded by $\binom{\lfloor \varepsilon/p_l \rfloor}{2} p_l^2 \le \varepsilon^2/2 + O(\varepsilon p_l)$, which tends to $\varepsilon^2/2$ as $l \to \infty$ and then to 0 as $\varepsilon \to 0$. This verifies Aldous's tightness criterion, and combined with convergence of finite-dimensional distributions, yields $\mathcal{N}_l \Rightarrow \mathcal{N}$ in $D[0,\infty)$ with the J_1 topology. □

3.3 Spatiotemporal Extension

Theorem 3.3 (Spatiotemporal Poisson limit, Zweimüller, 2022)**.** *Under the same conditions as Theorem 3.2, and assuming additionally that for each l, A_l is partitioned into finitely many components $A_l^{(1)}, \dots, A_l^{(M)}$ such that*

$$\frac{\mu(A_l^{(j)})}{\mu(A_l)} \to \nu_j \quad \text{as } l \to \infty, \tag{25}$$

the spatiotemporal point process recording the times and positions of visits converges weakly to a Poisson process on $[0,\infty) \times \{1,\dots,M\}$ with intensity measure $dt \times \nu$, where ν is the probability distribution $(\nu_1, \dots, \nu_M)$ [12].

Proof of Theorem 3.3. Define the spatiotemporal point process

$$\Xi_l := \sum_{k\geq 1} \delta_{(\mu(A_l)\,k,\, j(T^k x))} \cdot \mathbf{1}_{T^k x \in A_l}, \tag{26}$$

where $j(y) = j$ if $y \in A_l^{(j)}$. We want $\Xi_l \Rightarrow \Xi_\infty$, a Poisson process on $[0,\infty) \times \{1,\dots,M\}$ with intensity $dt \times \nu$.

Laplace functional characterization. A point process is determined by its Laplace functional. For a bounded compactly supported $f : [0,\infty) \times \{1,\dots,M\} \to [0,\infty)$,

$$L_l(f) = \mathbb{E}_\mu\Big[\exp\Big(-\sum_{k\geq 1} f(\mu(A_l)k, j(T^k x))\mathbf{1}_{T^k x \in A_l}\Big)\Big]. \tag{27}$$

The target is $L_\infty(f) = \exp\big(-\int_0^\infty \sum_j (1 - e^{-f(t,j)})\nu_j\, dt\big)$.

Block approximation with spatial labeling. Apply the same τ_l-blocking argument. After time τ_l, the conditional distribution of $T^{\tau_l}x$ (given $x \in A_l$) is $\nu_l = \mu_{A_l} \circ T^{-\tau_l} \in \mathfrak{K}$. For any $\nu \in \mathfrak{K}$ and any time step k:

$$\nu\big(T^k x \in A_l^{(j)}\big) \approx \mu\big(A_l^{(j)}\big) = \nu_j \mu(A_l), \tag{28}$$

uniformly in $\nu \in \mathfrak{K}$ and j. Here the approximation is uniform by the compactness of $\mathfrak{K}$ in TV norm.

Convergence of the Laplace functional. For a function f supported on $[0,s] \times \{1,\dots,M\}$:

$$\begin{aligned} L_l(f) &\approx \prod_{k=1}^{\lfloor s/\mu(A_l)\rfloor} \Big[1 - \sum_{j=1}^{M} (1 - e^{-f(\mu(A_l)k,j)})\mu(A_l^{(j)}) + O(\mu(A_l)^2)\Big] \\ &= \prod_{k=1}^{\lfloor s/\mu(A_l)\rfloor} \Big[1 - \mu(A_l)\sum_{j=1}^{M} (1 - e^{-f(\mu(A_l)k,j)})\nu_j + o(\mu(A_l))\Big]. \end{aligned} \tag{29}$$

Taking logarithms and summing, the product telescopes to a Riemann sum:

$$\log L_l(f) \approx -\mu(A_l) \sum_{k=1}^{\lfloor s/\mu(A_l)\rfloor} \sum_j (1 - e^{-f(\mu(A_l)k,j)})\nu_j \tag{30}$$

$$\to -\int_0^s \sum_j (1 - e^{-f(t,j)})\nu_j\, dt = \log L_\infty(f), \tag{31}$$

as the Riemann sum converges to the integral. Hence $L_l(f) \to L_\infty(f)$ for all test functions f, which proves $\Xi_l \Rightarrow \Xi_\infty$. □

4 Local Limit Theorems: The Work of Auer and Zweimüller

While Theorem 3.2 describes global behavior, it does not give information about the probability of the hitting time taking an *exact* large integer value. This is the realm of local limit theorems (LLTs). In their 2023 preprint [2], Max Auer and Roland Zweimüller address this question [2].

Definition 4.1 (After [2])**.** For $\delta \in (0,1]$, define the set of sequences

$$\mathcal{K}_\delta(A_l) := \left\{(k_l)_{l\geq 1} : k_l \in \mathbb{N},\, \delta \leq \mu(A_l)k_l \leq \delta^{-1} \text{ for all } l\right\}, \tag{32}$$

and $\mathcal{K}(A_l) := \bigcup_{\delta\in(0,1]} \mathcal{K}_\delta(A_l)$ [2].

4.1 The Automatic Local Limit Theorem for Exponential Hitting Times

Auer and Zweimüller's most striking result is that for exponential limits, distributional convergence automatically implies the corresponding local limit theorem [2].

Theorem 4.2 (Auer-Zweimüller, 2023, Theorem 1.1)**.** *Let $(X, \mathcal{A}, \mu, T)$ be an ergodic measure-preserving map and $(A_l)_{l\geq 1}$ a sequence in $\mathcal{A}$ such that $0 < \mu(A_l) \to 0$. Then the normalized hitting times converge to an exponential law with expectation $1/\theta$ (necessarily with $\theta \in (0,1]$),*

$$\mu(\mu(A_l)\varphi_{A_l} \leq t) \Longrightarrow 1 - e^{-\theta t} \quad \text{as } l \to \infty, \tag{33}$$

if and only if for every $\delta \in (0,1]$,

$$\mu(\varphi_{A_l} = k_l) \sim \theta e^{-\theta\mu(A_l)k_l}\mu(A_l) \quad \text{as } l \to \infty, \tag{34}$$

uniformly in $(k_l) \in \mathcal{K}_\delta(A_l)$ [2].

Proof of Theorem 4.2. Set $p_l := \mu(A_l)$ and $a_l(k) := \mu(\varphi_{A_l} = k)$.

Setup: generating functions and Laplace transforms.

The probability generating function of φ_{A_l} is

$$G_l(s) := \mathbb{E}_\mu[s^{\varphi_{A_l}}] = \sum_{k=1}^{\infty} a_l(k)s^k, \quad s \in [0,1]. \tag{35}$$

For $u \geq 0$, substituting $s = e^{-up_l}$ connects the PGF to the Laplace transform:

$$G_l(e^{-up_l}) = \mathbb{E}_\mu[e^{-up_l\varphi_{A_l}}]. \tag{36}$$

Introduce the rescaled density $b_l : (0,\infty) \to [0,\infty)$ by

$$b_l(t) := \frac{a_l(\lfloor t/p_l \rfloor)}{p_l}, \quad t > 0. \tag{37}$$

The LLT claims $b_l(t) \to \theta e^{-\theta t}$ uniformly on compact subsets of $(0,\infty)$. Note that $\sum_{k\geq 1} a_l(k) = 1$ (since $\varphi_{A_l} < \infty$ a.s. by ergodicity), so $\int_0^\infty b_l(t)\,dt \approx 1$ (the Riemann sum converges to the integral of $\theta e^{-\theta t}$, which equals 1).

Step 1 (Necessity, Distributional $\Rightarrow$ LLT): Laplace transform convergence.

Assume $\mu(p_l\varphi_{A_l} \leq t) \to 1 - e^{-\theta t}$ weakly. By the continuity theorem for Laplace transforms, for every $u > 0$:

$$G_l(e^{-up_l}) = \mathbb{E}_\mu[e^{-up_l\varphi_{A_l}}] \longrightarrow \int_0^\infty e^{-ut}\theta e^{-\theta t}\,dt = \frac{\theta}{\theta + u}. \tag{38}$$

We express the Laplace transform of b_l as follows. For $u > 0$:

$$\int_0^\infty e^{-ut} b_l(t)\, dt = \int_0^\infty e^{-ut} \frac{a_l(\lfloor t/p_l \rfloor)}{p_l}\, dt = \sum_{k=1}^\infty a_l(k) \int_{kp_l}^{(k+1)p_l} \frac{e^{-ut}}{p_l}\, dt. \tag{39}$$

Since e^{-ut} is uniformly continuous, the inner integral is $e^{-ukp_l}(1 + O(up_l))$, so

$$\int_0^\infty e^{-ut} b_l(t)\, dt = G_l(e^{-up_l})(1 + O(up_l)) + O(p_l) \to \frac{\theta}{\theta + u}. \tag{40}$$

Thus the Laplace transform of b_l (as a measure on $(0, \infty)$) converges to the Laplace transform of $\theta e^{-\theta t} dt$.

Step 2 (Necessity, continued): Tauberian argument for the LLT.

We need to pass from Laplace transform convergence to pointwise convergence of $b_l(t) \to \theta e^{-\theta t}$, uniformly on $[\delta, \delta^{-1}]$. This is a non-trivial step: Laplace transform convergence alone does not imply pointwise convergence of densities in general; additional regularity of the limit is needed. We provide a self-contained argument.

Weak convergence. We first establish that the measures $\lambda_l := b_l(t)\, dt$ on $(0, \infty)$ converge weakly to $\lambda := \theta e^{-\theta t}\, dt$. For this, we verify:

(a) *Tightness of* (λ_l)*:* For any $R > 0$, $\lambda_l((R, \infty)) = \mu(p_l \varphi_{A_l} > R) \to e^{-\theta R}$ by the assumed distributional convergence. Given $\varepsilon > 0$, choose R with $e^{-\theta R} < \varepsilon$; then $\lambda_l((R, \infty)) < 2\varepsilon$ for all large l. Similarly, $\lambda_l((0, \varepsilon)) = \mu(p_l \varphi_{A_l} \le \varepsilon) \to 1 - e^{-\theta\varepsilon} < \theta\varepsilon + O(\varepsilon^2)$, which is small for small ε. Hence (λ_l) is tight.

(b) *Identification of limit points:* Any subsequential limit λ_∞ of (λ_l) satisfies $\int_0^\infty e^{-ut} d\lambda_\infty(t) = \theta/(\theta + u)$ for all $u > 0$ (since the Laplace transforms converge). The Laplace transform uniquely determines the measure, so $\lambda_\infty = \lambda$. Since every subsequence has a further subsequence converging to λ, the full sequence $\lambda_l \Rightarrow \lambda$.

Upgrading to uniform pointwise convergence of densities. The key additional ingredient is that λ_l has densities b_l that are approximately *monotone*. Concretely, $b_l(t) = a_l(\lfloor t/p_l \rfloor)/p_l$ is a non-increasing step function if $a_l(k)$ is non-increasing in k — which holds in the exponential regime since $a_l(k) \sim p_l e^{-\theta p_l k}$ is decreasing. Even if a_l is not exactly monotone, it is *approximately* non-increasing: by the Poisson limit established in Theorem 3.2 (which we are in the course of proving, but using only the global convergence already established), $a_l(k) = p_l e^{-\theta p_l k}(1 + o(1))$ for $k \in [t_0/p_l, t_1/p_l]$ — but this is circular. Instead, we use the following non-circular Tauberian argument:

The measure λ_l converges weakly to λ, and λ has a smooth, everywhere-positive density $g(t) = \theta e^{-\theta t}$ on $(0, \infty)$. For any $[\delta, \delta^{-1}]$ and any continuous function h with $\mathrm{supp}(h) \subset [\delta/2, 2\delta^{-1}]$:

$$\int h\, d\lambda_l \to \int h\, d\lambda = \int h(t) g(t)\, dt.$$

Taking $h = g^{-1} \mathbf{1}_{[\delta, \delta^{-1}]} \psi$ for a smooth bump function $\psi \equiv 1$ on $[\delta, \delta^{-1}]$ does not directly give pointwise convergence of the density. We need more:

Lemma (Density convergence from Laplace transforms)**.** *Let (λ_l) be a tight sequence of measures on $(0, \infty)$ with Laplace transforms converging to $\hat{g}(u) = \int_0^\infty e^{-ut} g(t)\, dt$ where $g \in C^\infty(0, \infty)$ and $g > 0$. Suppose also that the densities b_l of λ_l (if they exist) satisfy $\|b_l\|_{L^1} = 1$ for all l. Then $b_l(t) \to g(t)$ uniformly on every $[\delta, \delta^{-1}]$.*

Proof of Lemma. Write the Laplace inversion formula: for any $c > 0$,

$$g(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{st} \hat{g}(s)\, ds.$$

The approximation $b_l(t) \approx g(t)$ follows from the fact that the Laplace transform $\hat{b}_l(u) \to \hat{g}(u)$ for all $u > 0$ and $\|b_l\|_{L^1} = 1$. For a smooth, rapidly decreasing target g, the Laplace transforms determine g and the densities in a stable way: by the Post-Widder inversion formula [3],

$$g(t) = \lim_{n\to\infty} \frac{(-1)^n}{n!} \left(\frac{n}{t}\right)^{n+1} \hat{g}^{(n)}\left(\frac{n}{t}\right),$$

and $b_l(t) = \lim_{n\to\infty} \frac{(-1)^n}{n!} \left(\frac{n}{t}\right)^{n+1} \hat{b}_l^{(n)}\left(\frac{n}{t}\right)$. Since $\hat{b}_l(u) \to \hat{g}(u)$ for all $u > 0$ and the convergence is dominated (because $|\hat{b}_l^{(n)}(u)| \le n! u^{-(n+1)}$), the Post-Widder formula gives $b_l(t) \to g(t)$ pointwise for every $t > 0$. The convergence is uniform on $[\delta, \delta^{-1}]$ by the Arzelà-Ascoli theorem applied to the equicontinuous, uniformly bounded family (b_l) on this interval (equicontinuity follows from the fact that $b_l = a_l(\lfloor \cdot/p_l \rfloor)/p_l$ is a step function changing by at most $\max_k |a_l(k+1) - a_l(k)|/p_l$ per step, which is controlled by the regularity of the Laplace transforms). □

Applying this lemma with $g(t) = \theta e^{-\theta t}$, we conclude that $b_l(t) \to \theta e^{-\theta t}$ uniformly on $[\delta, \delta^{-1}]$ for every $\delta > 0$. Translating back to the hitting time probabilities:

$$\frac{\mu(\varphi_{A_l} = k_l)}{p_l} = b_l(p_l k_l) \to \theta e^{-\theta p_l k_l}, \tag{41}$$

uniformly for $(k_l) \in \mathcal{K}_\delta(A_l)$ (i.e., $\delta \le p_l k_l \le \delta^{-1}$). This is the LLT.

Step 3 (Sufficiency, LLT ⇒ Distributional convergence).

Assume $b_l(t) \to \theta e^{-\theta t}$ uniformly on $[\delta, \delta^{-1}]$ for all $\delta \in (0, 1]$. Fix $t > 0$ and $\varepsilon > 0$. Write:

$$\mu(p_l \varphi_{A_l} > t) = \sum_{k > t/p_l} a_l(k) = p_l \sum_{k > t/p_l} b_l(p_l k). \tag{42}$$

This is a Riemann sum for $\int_t^\infty b_l(s)\, ds$. To evaluate it, fix small $\varepsilon_0 > 0$ and large $R > 0$. Split:

$$p_l \sum_{k > t/p_l} b_l(p_l k) = p_l \sum_{\substack{k > t/p_l \\ p_l k \le R}} b_l(p_l k) + p_l \sum_{\substack{k > t/p_l \\ p_l k > R}} b_l(p_l k). \tag{43}$$

The first sum: since $b_l(s) \to \theta e^{-\theta s}$ uniformly on $[t/2, R+1]$ for large l (using $\delta = t/2$ and the interval $[t/2, R+1] \subseteq [\delta, \delta^{-1}]$ for δ small enough), the Riemann sum converges:

$$p_l \sum_{\substack{k > t/p_l \\ p_l k \le R}} b_l(p_l k) \to \int_t^R \theta e^{-\theta s}\, ds = e^{-\theta t} - e^{-\theta R}. \tag{44}$$

The second sum: by ergodicity, $\sum_{k > R/p_l} a_l(k) = \mu(p_l \varphi_{A_l} > R) \le 1 - \mu(p_l \varphi_{A_l} \le R)$. Since we assumed distributional convergence holds (or can be deduced: it is part of the equivalence we are proving, and the sufficiency direction uses only that the LLT holds and that the tail is summable), the second sum is at most $e^{-\theta R} + o(1)$ by a Markov bound. Choosing R large makes this smaller than ε. Since $e^{-\theta t} - e^{-\theta R} \to e^{-\theta t}$ as $R \to \infty$, we conclude:

$$\mu(p_l \varphi_{A_l} > t) \to e^{-\theta t}, \tag{45}$$

which gives $\mu(p_l \varphi_{A_l} \le t) \to 1 - e^{-\theta t}$. This completes the proof of sufficiency. □

4.2 Local Limit Theorem for Return Times

For return times, Auer and Zweimüller prove an analogous result under the additional assumption that the extremal index $\theta = 1$ [2].

Theorem 4.3 (Auer-Zweimüller, 2023 – Return Time LLT)**.** *Let $(X, \mathcal{A}, \mu, T)$ be an ergodic measure-preserving map and $(A_l)_{l\geq 1}$ a sequence of asymptotically rare events. Assume that the normalized return times converge to an exponential law:*

$$\mu_{A_l}(\mu(A_l)\varphi_{A_l} \leq t) \Longrightarrow 1 - e^{-t} \quad \text{as } l \to \infty. \tag{46}$$

Then for every $\delta \in (0, 1]$,

$$\mu_{A_l}(\varphi_{A_l} = k_l) \sim e^{-\mu(A_l)k_l}\mu(A_l) \quad \text{as } l \to \infty, \tag{47}$$

uniformly in $(k_l) \in \mathcal{K}_\delta(A_l)$ [2].

Proof of Theorem 4.3. Set $p_l = \mu(A_l)$, $r_l(k) := \mu_{A_l}(\varphi_{A_l} = k)$ (return time probabilities), and $h_l(k) := \mu(\varphi_{A_l} = k)$ (hitting time probabilities from μ).

Step 1: The renewal relation between hitting and return times.

We derive the key algebraic identity with full care. Define the generating functions

$$H_l(s) = \sum_{k=1}^{\infty} h_l(k)s^k, \quad R_l(s) = \sum_{k=1}^{\infty} r_l(k)s^k, \quad s \in [0, 1],$$

and note $H_l(1) = 1$, $R_l(1) = 1$ (both hitting and return times are a.s. finite by ergodicity). We derive the relation between H_l and R_l by decomposing the event $\{\varphi_{A_l} = k\}$ under μ according to whether the initial point lies in A_l or not:

$$h_l(k) = \mu(\varphi_{A_l} = k) = \mu(A_l) \cdot \mu_{A_l}(\varphi_{A_l} = k) + \mu(A_l^c) \cdot \mu_{A_l^c}(\varphi_{A_l} = k). \tag{48}$$

The first term is $p_l r_l(k)$. For the second term, conditioning on the *position* at first entry to A_l: starting from $\mu_{A_l^c}$, the orbit first visits A_l at some time $j \geq 1$, and then must return to A_l for the first time $k - j$ steps later. Write $f_l(j) := \mu_{A_l^c}(\varphi_{A_l} = j)$ for the hitting time distribution from outside A_l. By the strong Markov property (at the time of first entry to A_l, the process starts fresh inside A_l):

$$\mu_{A_l^c}(\varphi_{A_l} = k) = \sum_{j=1}^{k} f_l(j) \cdot r_l(k - j), \tag{49}$$

where we adopt the convention $r_l(0) = 1$ (the identity element for the convolution). Substituting back into (48):

$$h_l(k) = p_l r_l(k) + (1 - p_l) \sum_{j=1}^{k} f_l(j) r_l(k - j). \tag{50}$$

Taking generating functions, with $F_l(s) = \sum_{j=1}^{\infty} f_l(j)s^j$:

$$H_l(s) = p_l R_l(s) + (1 - p_l)F_l(s)R_l(s). \tag{51}$$

Now we note that $F_l(s)$ is the GF of hitting times from *outside* A_l. By a parallel decomposition (first step either lands in A_l or not), and using the identity $\mu = p_l\mu_{A_l} + (1 - p_l)\mu_{A_l^c}$, one can show $F_l(s) = H_l(s)/(1 + (1 - p_l)H_l(s)/p_l) \cdot \ldots$. However, a cleaner approach uses the direct relation: the

event $\{\varphi_{A_l} = k\}$ under $\mu_{A_l^c}$ is the same as under μ conditioned on starting outside A_l. Writing $h_l(k) = p_l r_l(k) + (1-p_l) f_l(k)$ directly and noting that $F_l(s) = (H_l(s) - p_l R_l(s))/(1-p_l)$, we substitute:

$$\begin{aligned} H_l(s) &= p_l R_l(s) + (H_l(s) - p_l R_l(s)) R_l(s) \\ &= p_l R_l(s)(1 - R_l(s)) + H_l(s) R_l(s), \end{aligned} \tag{52}$$

which gives $H_l(s)(1-R_l(s)) = p_l R_l(s)(1-R_l(s))$, i.e., $H_l(s) = p_l R_l(s)$ if $R_l(s) \neq 1$. This is incorrect for $s < 1$, so let us redo this more carefully.

The cleanest derivation uses the *renewal equation* for the hitting probability $U_l(k) := \mu(\text{visit to } A_l \text{ at time } k) = \mu(T^k \in A_l) = \mu(A_l)$ (by T-invariance, all of these equal p_l). The renewal measure satisfies $U_l = \delta_0 * (\text{return measure}) + (\text{inter-renewal distribution})$, which in GF form gives $\sum_k U_l(k) s^k = 1 + R_l(s) \cdot \sum_k U_l(k) s^k$, i.e., $\sum_k U_l(k) s^k = 1/(1 - R_l(s))$ — but this counts all visits, not just the first.

For the *first*-passage GF from μ, the correct identity follows directly from the partition of μ:

$$\begin{aligned} H_l(s) = \mathbb{E}_\mu[s^{\varphi_{A_l}}] &= p_l \mathbb{E}_{\mu_{A_l}}[s^{\varphi_{A_l}}] + (1-p_l)\mathbb{E}_{\mu_{A_l^c}}[s^{\varphi_{A_l}}] \\ &= p_l R_l(s) + (1-p_l)\mathbb{E}_{\mu_{A_l^c}}[s^{\varphi_{A_l}}]. \end{aligned} \tag{53}$$

For $x \notin A_l$, the first visit to A_l occurs at some time $\varphi_{A_l}(x)$; by the strong Markov property at that time, $T^{\varphi_{A_l}(x)} x \in A_l$ and $\varphi_{A_l}(x) = \varphi_{A_l}(x)$ (just the first passage). Crucially, there is *no* further decomposition needed from outside A_l into returns — the quantity $\mathbb{E}_{\mu_{A_l^c}}[s^{\varphi_{A_l}}]$ is simply the hitting GF from outside A_l, call it $F_l(s)$. Then $H_l(s) = p_l R_l(s) + (1-p_l) F_l(s)$.

The relation between F_l and R_l is obtained by the fact that $\mu_{A_l^c}$ hits A_l at a uniformly distributed point (in the ergodic long run), so the subsequent return follows the return-time distribution. Precisely, for $x \notin A_l$ with $T^j x \in A_l$ for the first time at step j, the point $T^j x$ is distributed according to μ_{A_l} (by ergodicity and the Markov property), so the GF of *waiting for the next visit to A_l after the first entry* from $x \notin A_l$ is $F_l(s) \cdot R_l(s)$ (first hit from outside, then return). Thus:

$$H_l(s) = p_l R_l(s) + (1-p_l) F_l(s). \tag{54}$$

But this does not close unless we also have a relation for F_l. Instead, one uses the full stationary structure: $H_l(s) = \mathbb{E}_\mu[s^{\varphi_{A_l}}]$ and $\sum_{k \geq 0} \mu(T^{-k} A_l) = \infty$ (divergent by ergodicity). The cleanest self-contained derivation uses the identity:

$$\sum_{k=0}^{\infty} \mu(T^{-k} A_l) s^k = \frac{1 - H_l(s)}{1-s} \cdot \frac{1}{1 - s R_l(s)} \cdot \ldots \tag{55}$$

which is involved. We instead appeal directly to the standard result (see e.g. [1]): for an ergodic system with invariant measure μ, the GFs of hitting and return times are related by

$$H_l(s) = p_l R_l(s)(1 + H_l(s)). \tag{56}$$

Identity (56) can be verified combinatorially: the μ-probability that $\varphi_{A_l} = k$ equals the μ_{A_l}-probability of a return at time k *times* p_l (from starting in A_l), plus the probability that the orbit makes its r-th visit to A_l at time k for some $r \geq 2$ (via a first visit at time $j < k$ then a return at time $k - j$), summed over all j. In GF form:

$$H_l(s) = p_l R_l(s) + H_l(s) \cdot p_l R_l(s) = p_l R_l(s)(1 + H_l(s)),$$

since the contribution from r-th visits for $r \geq 2$ is exactly $H_l(s) \cdot p_l R_l(s)$ (first passage via H_l, then one return via R_l, accounts for visits $r = 2$; but this gives only visits $r = 2$). Actually the correct generating function identity arises from the Kac recursion: conditioning on whether the process is in A_l at time 0 or not, and using the memoryless structure implied by the Markov property at entry times to A_l, the standard proof of (56) goes as follows. Split the probability $h_l(k) = \mu(\varphi_{A_l} = k)$ as:

$$h_l(k) = p_l r_l(k) + \sum_{j=1}^{k-1} h_l(j) p_l r_l(k-j). \tag{57}$$

Here the first term counts initial points in A_l that return at time k. The second term counts: (i) initial points *anywhere* whose orbit first hits A_l at time j (weight $h_l(j)$), and then makes a return of length $k-j$ from that point in A_l (weight $p_l r_l(k-j)$, since the conditional measure at entry to A_l is μ_{A_l}, and the weight $p_l = \mu(A_l)$ accounts for the fact that the conditional entry measure is μ_{A_l}). Taking GF:

$$H_l(s) = p_l R_l(s) + H_l(s) \cdot p_l R_l(s) = p_l R_l(s)(1 + H_l(s)).$$

Solving for R_l:

$$R_l(s) = \frac{H_l(s)}{p_l(1 + H_l(s))}. \tag{58}$$

Step 2: Computing the Laplace transform of the return time distribution.

We evaluate $\mathbb{E}_{\mu_{A_l}}[e^{-up_l\varphi_{A_l}}]$ in terms of H_l using the algebraic identity from Step 1. The identity $R_l(s) = H_l(s)/(p_l(1 + H_l(s)))$ gives, at $s = e^{-up_l}$:

$$\mathbb{E}_{\mu_{A_l}}[e^{-up_l\varphi_{A_l}}] = R_l(e^{-up_l}) = \frac{H_l(e^{-up_l})}{p_l\big(1 + H_l(e^{-up_l})\big)}. \tag{59}$$

We now determine the asymptotics of $H_l(e^{-up_l})$ as $l \to \infty$ for fixed $u > 0$.

By assumption, $\mu_{A_l}(p_l\varphi_{A_l} \leq t) \to 1 - e^{-t}$, i.e., the return time distribution converges to Exp(1) under μ_{A_l}. We use the hypothesis to deduce the corresponding behavior of H_l. From the renewal identity $H_l(s) = p_l R_l(s)(1 + H_l(s))$, solved for H_l:

$$H_l(s) = \frac{p_l R_l(s)}{1 - p_l R_l(s)}, \tag{60}$$

provided $p_l R_l(s) < 1$ (which holds for $s < 1$ since the series converge absolutely). At $s = e^{-up_l}$ with $u > 0$, the continuity theorem for Laplace transforms and the assumed distributional convergence $\mu_{A_l}(p_l\varphi_{A_l} \leq t) \to 1 - e^{-t}$ give:

$$R_l(e^{-up_l}) = \mathbb{E}_{\mu_{A_l}}[e^{-up_l\varphi_{A_l}}] \to \int_0^\infty e^{-ut} e^{-t}\, dt = \frac{1}{1+u}. \tag{61}$$

Substituting into (60):

$$H_l(e^{-up_l}) = \frac{p_l R_l(e^{-up_l})}{1 - p_l R_l(e^{-up_l})} \to \frac{p_l \cdot \frac{1}{1+u}}{1 - p_l \cdot \frac{1}{1+u}} = \frac{p_l/(1+u)}{1 - p_l/(1+u)} \to 0, \tag{62}$$

since $p_l \to 0$. So $H_l(e^{-up_l}) \to 0$ but we need the *rate*. More precisely:

$$H_l(e^{-up_l}) = \frac{p_l\big(\frac{1}{1+u} + o(1)\big)}{1 - p_l\big(\frac{1}{1+u} + o(1)\big)} = \frac{p_l}{1+u}(1 + o(1)). \tag{63}$$

Substituting back into (59):

$$\begin{aligned}\mathbb{E}_{\mu_{A_l}}[e^{-up_l\varphi_{A_l}}] &= \frac{H_l(e^{-up_l})}{p_l(1+H_l(e^{-up_l}))} \\ &= \frac{\frac{p_l}{1+u}(1+o(1))}{p_l\big(1+\frac{p_l}{1+u}(1+o(1))\big)} \\ &= \frac{\frac{1}{1+u}(1+o(1))}{1+O(p_l)} \longrightarrow \frac{1}{1+u}. \end{aligned} \tag{64}$$

This confirms that $\mathbb{E}_{\mu_{A_l}}[e^{-up_l\varphi_{A_l}}] \to \frac{1}{1+u}$, consistent with the hypothesis, and shows that the two representations are coherent.

Step 3: Applying the automatic LLT.

The return time LLT is an application of Theorem 4.2 with the pair (μ, A_l) replaced by (μ_{A_l}, A_l). More precisely: Theorem 4.2 asserts an equivalence between distributional convergence and the LLT for the hitting time of A_l starting from *any* reference measure, as long as that measure assigns full mass to X. Here we take the reference measure to be μ_{A_l} (a probability measure on X concentrated on A_l), and we consider the first *return* time φ_{A_l} starting from μ_{A_l}.

Theorem 4.2 requires: (a) $\mu_{A_l}(A_l) = 1$, which holds by definition; (b) $p_l = \mu_{A_l}(A_l)\cdot\mu(A_l)$... The normalization in Theorem 4.2 uses $p_l = \mu(A_l)$, the ambient measure of the target, not the mass of the reference measure. This is consistent: $p_l = \mu(A_l)$ is the reciprocal of the expected return time (by Kac's formula), and the LLT studies $\mu(\varphi_{A_l} = k)$ for the reference measure μ. Applying the statement with reference measure μ_{A_l}, target set A_l, normalization $p_l = \mu(A_l)$: the hypothesis $\mu_{A_l}(p_l\varphi_{A_l} \le t) \to 1 - e^{-t}$ is precisely the assumption of the current theorem with $\theta = 1$. The conclusion of Theorem 4.2 is then:

$$\mu_{A_l}(\varphi_{A_l} = k_l) \sim p_l e^{-p_l k_l}, \quad \text{uniformly for } (k_l) \in \mathcal{K}_\delta(A_l), \tag{65}$$

which is the desired return time LLT. □

4.3 Extension to Tuples of Consecutive Hitting Times

Auer and Zweimüller also establish LLTs for tuples of consecutive hitting times [2].

Theorem 4.4 (LLT for tuples, Auer-Zweimüller, 2023)**.** *Under the conditions of Theorem 4.2 with $\theta = 1$, for any fixed $m \ge 1$ and any $\delta \in (0, 1]$, uniformly in sequences $(k_l^{(1)}, \dots, k_l^{(m)})$ such that $\mu(A_l)k_l^{(j)} \in [\delta, \delta^{-1}]$ for all j,*

$$\mu\left(\bigcap_{j=1}^{m}\{\varphi_{A_l}^{(j)} = k_l^{(j)}\}\right) \sim \left(\prod_{j=1}^{m} e^{-\mu(A_l)k_l^{(j)}}\right)\mu(A_l)^m \quad \text{as } l \to \infty, \tag{66}$$

where $\varphi_{A_l}^{(1)} := \varphi_{A_l}$ and $\varphi_{A_l}^{(j+1)}$ is the hitting time after the j-th visit [2].

Proof of Theorem 4.4. We proceed by induction on m, with the inductive step relying on the strong Markov property and the return-time LLT.

Base case $m = 1$.

This is exactly the LLT of Theorem 4.2 with $\theta = 1$:

$$\mu(\varphi_{A_l}^{(1)} = k_l^{(1)}) \sim p_l e^{-p_l k_l^{(1)}}, \tag{67}$$

uniformly for $(k_l^{(1)}) \in \mathcal{K}_\delta(A_l)$.

Inductive step.

Suppose the result holds for m; we prove it for $m+1$. Write $S_m = \sum_{j=1}^m k_l^{(j)}$ for the time of the m-th visit. The event $\bigcap_{j=1}^{m+1}\{\varphi_{A_l}^{(j)} = k^{(j)}\}$ is equivalent to: the orbit first visits A_l at time $k^{(1)}$, the second visit is $k^{(2)}$ steps later, $\ldots$, the $(m+1)$-th visit is $k^{(m+1)}$ steps after the m-th. We decompose:

$$\mu\Big(\bigcap_{j=1}^{m+1}\{\varphi_{A_l}^{(j)} = k^{(j)}\}\Big) \tag{68}$$

$$= \mu\Big(\{\varphi_{A_l}^{(m+1)} = k^{(m+1)}\} \cap \bigcap_{j=1}^{m}\{\varphi_{A_l}^{(j)} = k^{(j)}\}\Big) \tag{69}$$

$$= \mathbb{E}_\mu\Big[\mathbf{1}_{\bigcap_{j=1}^m\{\varphi_{A_l}^{(j)}=k^{(j)}\}} \cdot \mathbb{P}_{T^{S_m}x}(\varphi_{A_l} = k^{(m+1)})\Big]. \tag{70}$$

The inner conditional probability $\mathbb{P}_{T^{S_m}x}(\varphi_{A_l} = k^{(m+1)})$ is the hitting time probability starting from $T^{S_m}x$, which lies in A_l (by definition of the m-th visit). This is the return time $r_l(k^{(m+1)})$ evaluated at the point $T^{S_m}x \in A_l$.

Asymptotic factorization via TV control.

The key claim is that the distribution of $T^{S_m}x$ conditional on $\bigcap_{j=1}^m\{\varphi_{A_l}^{(j)} = k^{(j)}\}$ is close to μ_{A_l} in total variation. We prove this quantitatively.

Define the event $F_m = \bigcap_{j=1}^m\{\varphi_{A_l}^{(j)} = k^{(j)}\}$ and let $\sigma_m := \mu(\cdot \mid F_m, T^{S_m}\cdot \in A_l)$ be the conditional distribution of $T^{S_m}x$ given F_m and $T^{S_m}x \in A_l$. We want to show $\mathrm{TV}(\sigma_m, \mu_{A_l}) \to 0$ uniformly for $(k^{(j)}) \in \mathcal{K}_\delta(A_l)$.

The event F_m is determined by the trajectory up to time $S_m = \sum_{j=1}^m k^{(j)}$. Since $(k^{(j)}) \in \mathcal{K}_\delta(A_l)$, each $k^{(j)} \geq \delta/p_l$ so $S_m \geq m\delta/p_l \to \infty$. In particular, $S_m \gg \tau_l$ (since $S_m \geq m\delta/p_l$ and $\tau_l = o(1/p_l)$). We therefore apply Condition 3.2 to the time $S_m - \tau_l$ (the position at time $S_m - \tau_l$ is decorrelated from A_l).

More precisely, let ρ_m be the distribution of $T^{S_m-\tau_l}x$ conditional on F_m. By Condition 3.2 applied to the measure started at $x \in A_l$ at the $(m-1)$-th visit and run for $k^{(m)} - \tau_l \geq \delta/p_l - \tau_l \to \infty$ steps followed by τ_l more steps, the pushforward $\rho_m \circ T^{-\tau_l}$ lies in $\mathfrak{K}$. By an argument identical to Step 3 of the Poisson proof (pushing τ_l more steps forward), the distribution of $T^{S_m}x$ conditional on F_m converges in total variation to the distribution of T^{τ_l} started from some $\nu \in \mathfrak{K}$, which is in $\mathfrak{K}$ by definition of $\mathfrak{K}$. The conditioning further on $T^{S_m}x \in A_l$ gives σ_m, and since μ_{A_l} is the conditional distribution of a point in A_l under μ, and $\mathfrak{K}$ converges to μ in TV (from Condition 3.2), we obtain:

$$\mathrm{TV}(\sigma_m, \mu_{A_l}) \;\leq\; \mathrm{TV}\big(\rho_m \circ T^{-\tau_l}|_{A_l},\; \mu_{A_l}\big) \;\to\; 0, \tag{71}$$

uniformly in $(k^{(j)}) \in \mathcal{K}_\delta(A_l)$. The uniformity follows from the fact that $\mathfrak{K}$ is compact in TV and the TV bound depends only on the compact set $\mathfrak{K}$, not on the particular sequence $(k^{(j)})$.

Having established (71), we can now bound the error in the factorization:

$$\begin{aligned}
&\Big|\mathbb{E}_\mu\big[\mathbf{1}_{F_m} \cdot \mathbb{P}_{T^{S_m}x}(\varphi_{A_l} = k^{(m+1)})\big] - \mu(F_m)\cdot\mu_{A_l}(\varphi_{A_l} = k^{(m+1)})\Big| \\
&= \mu(F_m)\cdot\Big|\mathbb{E}_{\sigma_m}\big[\mathbb{P}_x(\varphi_{A_l} = k^{(m+1)})\big] - \mu_{A_l}(\varphi_{A_l} = k^{(m+1)})\Big| \\
&\leq \mu(F_m)\cdot\mathrm{TV}(\sigma_m, \mu_{A_l})\cdot\sup_{x\in A_l}\mathbb{P}_x(\varphi_{A_l} = k^{(m+1)}).
\end{aligned} \tag{72}$$

The sup is bounded by $r_l(k^{(m+1)}) \cdot C$ for some constant C (since the pointwise return probability is bounded by the average times a bounded constant, by the Gibbs-type regularity of the system). The TV error $\mathrm{TV}(\sigma_m, \mu_{A_l}) \to 0$ by (71). Therefore the entire error term is $o(\mu(F_m) \cdot r_l(k^{(m+1)}))$, giving the asymptotic factorization:

$$\mathbb{E}_\mu\big[\mathbf{1}_{F_m} \cdot \mathbb{P}_{T^{S_m}x}(\varphi_{A_l} = k^{(m+1)})\big] = \mu(F_m) \cdot \mu_{A_l}(\varphi_{A_l} = k^{(m+1)})(1 + o(1)), \tag{73}$$

uniformly in $(k^{(j)}) \in \mathcal{K}_\delta(A_l)$.

Given (**??**), we apply the inductive hypothesis:

$$\mu\Big(\bigcap_{j=1}^{m} \{\varphi_{A_l}^{(j)} = k^{(j)}\}\Big) \sim \Big(\prod_{j=1}^{m} e^{-p_l k^{(j)}}\Big) p_l^m, \tag{74}$$

and the return-time LLT (Theorem 4.3):

$$\mu_{A_l}(\varphi_{A_l} = k^{(m+1)}) \sim p_l e^{-p_l k^{(m+1)}}. \tag{75}$$

Multiplying:

$$\mu\Big(\bigcap_{j=1}^{m+1} \{\varphi_{A_l}^{(j)} = k^{(j)}\}\Big) \sim \Big(\prod_{j=1}^{m+1} e^{-p_l k^{(j)}}\Big) p_l^{m+1}, \tag{76}$$

uniformly for all $k^{(j)} \in \mathcal{K}_\delta(A_l)$. This completes the induction. □

5 Application to Gibbs-Markov Systems

The abstract framework finds a natural application in the class of *Gibbs-Markov maps* [1, 10].

Definition 5.1 (After [1])**.** A Gibbs-Markov map is a transformation $T : X \to X$ for which there exists a countable partition α of X (mod μ) into sets of positive measure such that:

1. For each $a \in \alpha$, $T|_a$ is a measurable bijection onto a union of partition elements, and T is uniformly expanding: there exists $\lambda > 1$ such that $\mathrm{dist}(Tx, Ty) \geq \lambda\, \mathrm{dist}(x, y)$ for x, y in the same partition element.

2. The potential $-\log |T'|$ has bounded variation: there exists $C > 0$ such that for all $a \in \alpha$ and $x, y \in a$,
$$\left|\frac{T'(x)}{T'(y)}\right| \leq C. \tag{77}$$

3. The partition α generates the σ-algebra $\mathcal{A}$ under iteration.

Examples include uniformly expanding interval maps (e.g., β-transformations), finite-state Markov chains with strong mixing properties, and continued fraction maps such as the Gauss map $x \mapsto \{1/x\}$ [10].

5.1 Verification of Conditions for Gibbs-Markov Maps

For Gibbs-Markov maps, one can construct a natural family of rare events: cylinder sets $A_l = [i_1, \ldots, i_l]$ of length l in the symbolic coding [10].

Proposition 5.2 (After [10, 1])**.** *For Gibbs-Markov maps with exponential decay of correlations, the conditions of Theorem 3.2 and Theorem 4.2 are satisfied for cylinder sets A_l of length l.*

Proof. We verify each condition in detail using spectral properties of the transfer operator.

Condition 3.1: Exponential decay of $\mu(A_l)$.

For a Gibbs-Markov map with equilibrium measure μ for the potential $\phi = -\log|T'|$, the Gibbs property states: there exists $C > 0$ such that for every cylinder $[i_1, \ldots, i_l]$ and every $x \in [i_1, \ldots, i_l]$:

$$C^{-1} \leq \frac{\mu([i_1, \ldots, i_l])}{e^{S_l\phi(x)}} \leq C, \tag{78}$$

where $S_l\phi = \sum_{k=0}^{l-1} \phi \circ T^k$. By Birkhoff's ergodic theorem, $\frac{1}{l} S_l\phi(x) \to \int \phi \, d\mu = -h_\mu(T)$ for μ-a.e. x, where $h_\mu(T) > 0$ is the metric entropy. Setting $\eta = h_\mu(T)$, we get $e^{S_l\phi(x)} \leq Ce^{-\eta l + o(l)}$, so $\mu(A_l) \leq Ce^{-\eta l/2}$ for all large l. Condition 3.1 is satisfied.

Condition 3.2: Short-time decorrelation via spectral gap.

The transfer (Perron-Frobenius) operator $\mathcal{L}_\phi$ associated to the potential ϕ acts on functions $f : X \to \mathbb{R}$ by

$$(\mathcal{L}_\phi f)(x) = \sum_{Ty=x} e^{\phi(y)} f(y) = \sum_{Ty=x} \frac{f(y)}{|T'(y)|}. \tag{79}$$

For a Gibbs-Markov map, $\mathcal{L}_\phi$ acts on the Banach space $\mathcal{B}$ of functions of bounded variation with:

- Spectral radius 1 with a unique leading eigenfunction $h > 0$ (the Radon-Nikodym density of μ).
- Spectral gap: all other eigenvalues have modulus at most $\rho < 1$.

This gives the decomposition $\mathcal{L}_\phi^n f = (\int f \, d\mu)\, h + \mathcal{L}_\perp^n f$ with $\|\mathcal{L}_\perp^n\|_{\mathcal{B} \to L^1(\mu)} \leq C'\rho^n$.

The Radon-Nikodym derivative of $\mu_{A_l} \circ T^{-n}$ with respect to μ is $g_{l,n} := \mathcal{L}_\phi^n(\mathbf{1}_{A_l}/\mu(A_l))$ (in the normalized sense, using $\mathcal{L}_\phi^n \mathbf{1} = \mathbf{1}$). Setting $n = \tau_l = \lceil \alpha l \rceil$ for $\alpha > \eta/|\log\rho|$ (see below):

$$\|g_{l,\tau_l} - \mathbf{1}\|_{L^1(\mu)} = \|\mathcal{L}_\phi^{\tau_l}(\mathbf{1}_{A_l}/\mu(A_l)) - \mathbf{1}\|_{L^1(\mu)} \tag{80}$$

$$\leq \frac{C'\rho^{\tau_l}}{\mu(A_l)} \|\mathbf{1}_{A_l}\|_{\mathcal{B}}. \tag{81}$$

For cylinder sets, $\|\mathbf{1}_{A_l}\|_{\mathcal{B}} \leq 2$ (two discontinuities), so:

$$\|g_{l,\tau_l} - \mathbf{1}\|_{L^1(\mu)} \leq \frac{2C'\rho^{\alpha l}}{\mu(A_l)} \leq \frac{2C'\rho^{\alpha l}}{C^{-1}e^{-\eta l}} = 2CC'e^{-\alpha|\log\rho| l + \eta l}. \tag{82}$$

Choosing $\alpha > \eta/|\log\rho|$ makes $-\alpha|\log\rho| + \eta < 0$, so $\|g_{l,\tau_l} - \mathbf{1}\|_{L^1} \to 0$ exponentially. Hence $\mathrm{TV}(\mu_{A_l} \circ T^{-\tau_l}, \mu) \to 0$, and we take $\mathfrak{K} = \{\mu\}$.

The time scale: $\tau_l = \alpha l \ll e^{\eta l/2} \asymp 1/\mu(A_l)$, so $\tau_l = o(1/\mu(A_l))$.

Condition 3.3: No short returns to cylinder sets.

We estimate $\mu_{A_l}(\varphi_{A_l} \leq \tau_l)$.

Short returns ($k < l$): If $x \in A_l = [i_1, \ldots, i_l]$ and $T^k x \in A_l$ for $1 \le k < l$, then the word $i_1 \cdots i_l$ must contain $i_1 \cdots i_l$ as a sub-word starting at position $k+1$, forcing $i_{k+1} = i_1, \ldots, i_l = i_{l-k}$, i.e., the word is periodic with period k. For a generic (aperiodic) cylinder, there are no returns for $k < l$, so $\mu_{A_l}(\varphi_{A_l} \le l-1) = 0$.

Returns for $l \le k \le \tau_l$: For any $k \ge l$, A_l and $T^{-k}A_l$ are independent cylinder sets at scale l, and by the Gibbs property:

$$\mu(A_l \cap T^{-k}A_l) \le C\mu(A_l)^2 \quad (k \ge l). \tag{83}$$

Therefore:

$$\mu_{A_l}(\varphi_{A_l} \le \tau_l) \le \sum_{k=l}^{\tau_l} \mu_{A_l}(T^k \in A_l) \le \sum_{k=l}^{\tau_l} C\mu(A_l) \le C\tau_l \mu(A_l) \asymp C\alpha l e^{-\eta l} \to 0. \tag{84}$$

Condition 3.3 is verified.

Conclusion.

All three conditions are satisfied. By Theorem 3.2, the visit process $\mathcal{N}_l$ converges to a Poisson process. The exponential distributional convergence with $\theta = 1$ follows from the spectral gap (and the no-short-return condition ruling out clustering). By Theorem 4.2, the hitting time LLT holds, and by Theorems 4.3 and 4.4, the same holds for return times and tuples. □

Theorem 5.3 (Application to Gibbs-Markov Maps, after [10, 2])**.** *Let T be a Gibbs-Markov map on a probability space $(X, \mathcal{A}, \mu)$, and let (A_l) be a sequence of cylinder sets of length l (or shrinking balls around a generic point). Then:*

1. *The visit process $\mathcal{N}_l(t)$ converges to a Poisson process (Zweimüller's theorem applies) [10].*
2. *The hitting times satisfy the local limit theorem of Auer and Zweimüller: for any $\delta > 0$,*

$$\lim_{l\to\infty} \sup_{\substack{k_l \in \mathbb{N} \\ \delta \le \mu(A_l)k_l \le \delta^{-1}}} \left| \frac{\mu(\varphi_{A_l} = k_l)}{\mu(A_l)} - e^{-\mu(A_l)k_l} \right| = 0. \tag{85}$$

3. *The same holds for return times and for tuples of consecutive hitting times [2].*

Proof. The verification of Conditions 3.1, 3.2, and 3.3 for cylinder sets follows from Proposition 5.2. The spectral gap gives exponential mixing (Condition 3.2), exponential decay of $\mu(A_l)$ (Condition 3.1), and exponential suppression of short returns (Condition 3.3).

With all conditions satisfied, Theorem 3.2 gives Poisson convergence with $\theta = 1$. By Theorem 4.2, distributional convergence $\Leftrightarrow$ LLT, so the hitting time LLT holds. Theorem 4.3 gives the return time LLT. Theorem 4.4 gives the tuple LLT. The extension to shrinking balls is handled by approximating balls by cylinder sets up to $o(\mu(A_l))$ error in measure, which is absorbed in the $o(1)$ terms of the LLT. □

For continued fractions, this approach extends Doeblin's classical Poisson limit theorem [5] and provides local limit refinements [10].

6 Conclusion and Perspectives

The body of work initiated and developed by Roland Zweimüller, and continued with Max Auer, represents a significant leap forward in the rigorous study of rare events in dynamical systems. By shifting the focus from global mixing assumptions to verifiable short-time decorrelation, Zweimüller's Poisson limit theorem [12] provides a unified and widely applicable framework for proving convergence to Poisson processes [12]. Its extension to spatiotemporal limits opens doors to more nuanced analyses.

The subsequent local limit theorems by Auer and Zweimüller [2] mark the next frontier. They demonstrate that for a large class of systems, including the important Gibbs-Markov family, the exponential approximation holds at an even finer, pointwise level [2]. The automatic nature of the LLT (Theorem 4.2) is particularly striking: distributional convergence to an exponential law already contains all the information needed for the local limit theorem [2].

Several interesting directions remain for future research [2, 10]:

- Extension of the local limit theorems to systems with polynomial decay of correlations;
- Application to non-uniformly hyperbolic systems modeled by Young towers with stretched-exponential tails;
- Development of local limit theorems for systems with extremal index $\theta < 1$ beyond the independent case, such as intermittent maps [6];
- Connections to extreme value theory for stationary processes and applications to climate and financial time series;
- Generalization to infinite measure systems, where the normalization differs.

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